

Day 58 - Wireless Configuration

Wednesday, September 9, 2026

8:38 PM

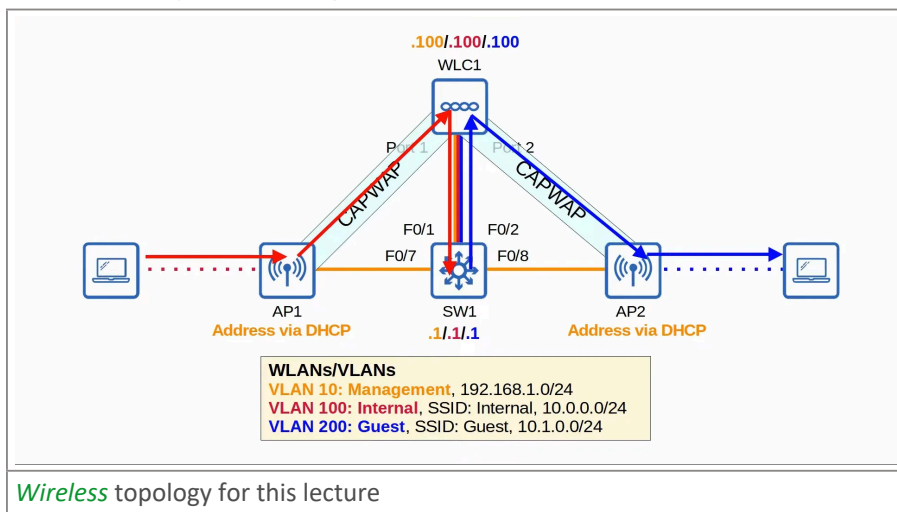
Wireless Configuration Info

There are many steps to configuring a *wireless* network on a network. In this lecture, we'll be setting up a *wireless* network with (1) WLC, (2) APs, and (1) Multilayer Switch.

- The **WLC** connects to the **Multilayer Switch** via a **Link Aggregation Group (LAG)**, which is **EtherChannel**.
- The **Multilayer Switch** will be running as the **DHCP & NTP** server. Which will provide the **APs** with addresses and their timezone.

Watch out: WLC and EtherChannels

- WLCs only support **static LAG (EtherChannel)**.
- Therefore, they can't run **PAgP** or **LACP**.



WLC Ports & Interfaces

WLC **ports** are the physical ports that cables connect to.

WLC **interfaces** are the logical interfaces within the WLC (ie. SVIs on a switch)

- WLCs have a few different kinds of **ports**:

1. Service Port

- A dedicated management port.
- Used for *out-of-band* management.
- Must connect to a switch *access* port because it only supports one VLAN.
- This port can be used to connect to the device while it is booting, perform system recovery etc.

2. Redundancy Port

- This port is used to connect to another **WLC** to form a *high availability (HA)* pair.

3. Distribution System Port

- These are the standard network ports that connect to the 'distribution system' (*wired* network) and are used for data traffic.
- These ports usually connect to switch *trunk* ports, and if multiple **distribution ports** are used, they can form a **LAG**.

4. Console Port

- This is a standard console port, either RJ45 or USB.

- WLCs also have a few different kind of **interfaces**:

1. Service Port Interface

- If the **service port** is used, the interface is bound to it and used for *out-of-band* management.

2. Redundancy Management Interface

- When the two **WLCs** are connected by their **redundancy ports**, one **WLC** is 'active' and the other is 'standby'.
- This interface can be used to connect to and manage the 'standby' **WLC**.
- 3. **Management Interface**
 - Used for management traffic such as **Telnet**, **SSH**, **HTTP**, **HTTPS**, **RADIUS** authentication, **NTP**, **Syslog**, etc.
 - **CAPWAP** tunnels are also formed to/from the **WLC's management interface**.
- 4. **Virtual Interface**
 - This interface is used when communicating with **wireless** clients to relay **DHCP** requests, perform web authentication, etc.
- 5. **Dynamic Interface**
 - These are the interfaces used to map a **WLAN** to a VLAN.
 - For example, traffic from the 'internal' **WLAN** will be sent to the **wired** network from the **WLC's** 'internal' **dynamic interface**.

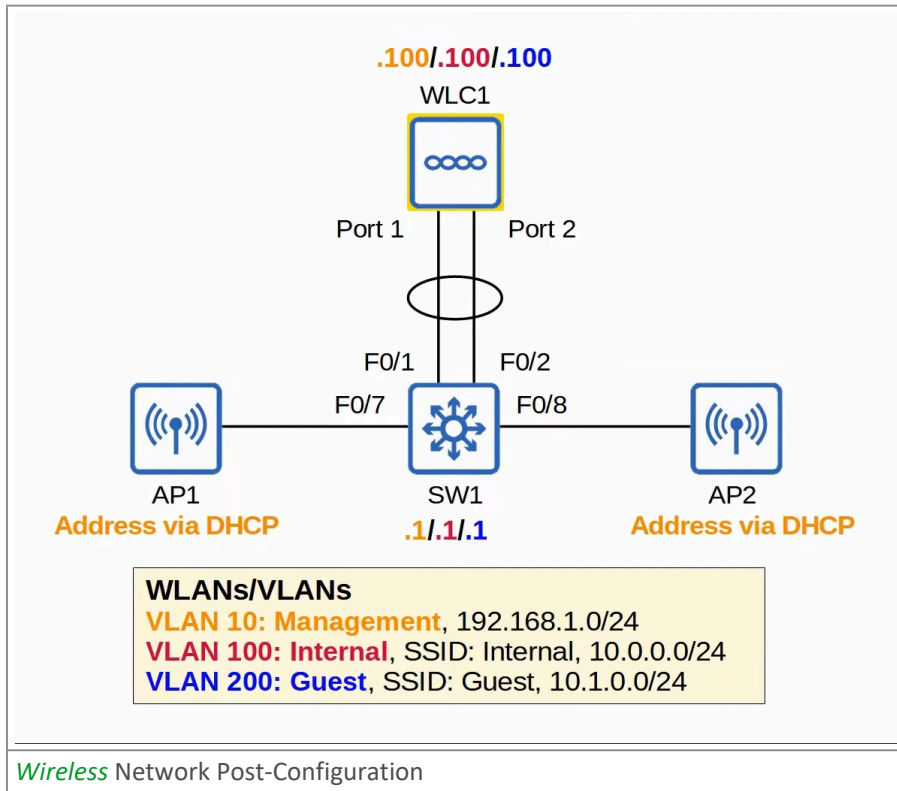
Multilayer Switch Configuration

- **VLAN**
 - In this step, we'll initialize the VLANs that we'll use for management, internal use, and guests.
 - `vlan 10`
 - `name Management`
 - `vlan 100`
 - `name Internal`
 - `vlan 200`
 - `name Guest`
- **Interface**
 - In this step, we'll setup the switchports to use the previously configured VLANs as well as set up **SVIs** for end-hosts to use as their default gateway.
 - `int range f0/6 - 8`
 - `switchport mode access`
 - `switchport access vlan 10`
 - `spanning-tree portfast`
 - `interface range f0/1 - 2`
 - `channel-group 1 mode on`
 - `interface port-channel 1`
 - `switchport mode trunk`
 - `switchport trunk allowed vlan 10,100,200`
 - `interface vlan 10`
 - `ip address 192.168.1.1 255.255.255.0`
 - `interface vlan 100`
 - `ip address 10.0.0.1 255.255.255.0`
 - `interface vlan 200`
 - `ip address 10.1.0.1 255.255.255.0`
- **DHCP**
 - **DHCP** configurations so that end-hosts are automatically assigned IP addresses
 - `ip dhcp pool VLAN10`
 - `network 192.168.1.0 255.255.255.0`
 - `default-router 192.168.1.1`
 - `option 43 ip 192.168.1.100`
 - `ip dhcp pool VLAN100`
 - `network 10.0.0.0 255.255.255.0`
 - `default-router 10.0.0.1`
 - `ip dhcp pool VLAN200`
 - `network 10.1.0.0 255.255.255.0`
 - `default-router 10.1.0.1`
- **NTP**
 - This step is to configure the switch as an **NTP** master in order to provide the time for end-hosts

- *ntp master*

What is **Option 43**?

- **Option 43** can be used to tell the **APs** the IP address of their **WLC**
- (This is not necessary in this case, because the **APs** and the **WLC** are in the same subnet. The **WLC** will hear the **APs** broadcast **CAPWAP discovery** messages)



WLC Configuration

1. WLC - Initial Setup

When booting into the WLC for the first time:

Step 1

- You'll be asked whether or not you want to terminate *autoinstall*, which can be used to automatically download the **WLC's** configuration from a **TFTP** server.
- We'll pick 'Yes' which terminates the *autoinstall* and allows us to manually configure the **WLC**.

Step 2

- After that, you'll be asked to set a *system name* for the **WLC** as well as configure a *username/password* to manage the **WLC** later on.

Step 3

- You'll get a prompt asking whether or not you'd like to enable **Link Aggregation Protocol (LAG)**.
- We'll pick 'Yes'

Step 4

- Then, we'll have to setup the management interface configurations. This includes: ^wlc-initial-management
 1. Management Interface IP Address - **192.186.1.100**
 2. Management Interface Netmask - **255.255.255.0**
 3. Management Interface Default Router - **192.168.1.1**
 4. Management Interface VLAN Identifier - **10**
 5. Management Interface DHCP Server IP Address - **192.168.1.1**

Step 5

- There are some extra options that you don't need to fully understand for the CCNA, but they still have to be configured, they are:
 1. Virtual Gateway IP Address - **172.16.1.1**

- Used when the **WLC** is communicating directly with wireless clients, while relaying **DHCP** messages for example
- 2. Multicast IP Address - **239.239.239.239**
- 3. Mobility/RF Group Name - **jITlab**

Step 6

- Afterwards, we are prompted for more configuration settings, which ask for:
 1. Network Name (**SSID**) - **Internal**
 2. Configure **DHCP** Bridging Mode - **No**
 - Set to '**No**' by default. If enabled, this will make the **WLC** transparent in **DHCP** communications between the clients and **DHCP** server, which makes it look like the clients are communicating directly with the **DHCP** server.
 3. Allow Static IP Addresses - **Yes**
 - Allows clients to choose a static IP address if they want prefer.
 4. Configure a **RADIUS** Server now? - **No**
 - The default **WLAN** security policy requires us to setup a **RADIUS** server.
 - This is not necessary for the CCNA, however, you do need to know how to set up a **Pre-Shared Key (PSK)**, which will be covered in the next section.
 5. Enter Country Code list - **FR**

Step 7

- Lastly, you'll have to configure the last settings of the configuration wizard, which are:
 1. Enable 802.11b Network? - **Default**
 2. Enable 802.11a Network? - **Default**
 3. Enable 802.11g Network? - **Default**
 4. Enable Auto-RF? - **Default**
 5. Configure a **NTP** server now?
 - We'll pick '**Yes**'
 - This will ask us to enter the **NTP** server's IP address (**192.168.1.1**) and pick a **polling interval**, we'll choose (**3600**).

2. WLC - Accessing the GUI

After completing the initial setup, we can now access the WLC's management page.

- To do this, we have to use a browser on a device and enter the **Management Interface IP Address** (**192.168.1.100**) that we configured earlier.
- This will open up the **Monitor** page by default which shows us general information about the **WLC**.

2.1 Controller Tab

In here, we can configure the **logical interfaces** that will later be connected to our **physical ports** in the WLC WLANs tab.

- To create a **logical interface**, we press on the 'New' button in the top right.
 - This will open a new page where we can select the **Interface Name** and **VLAN ID**.
 - Pressing 'Next' will lead us to the next page where we can assign the following:
 1. **VLAN Identifier**
 2. **IP Address**
 3. **Netmask**
 4. **Default Gateway**
 5. **Primary DHCP Server**

Controller Tab - Lab Configuration

For our current lab, we'll be setting up two interfaces:

1. 'Internal' **interface** - **VLAN ID** (**100**)
 - **VLAN Identifier** = **100**
 - **IP Address** = **10.0.0.100**
 - **Netmask** = **255.255.255.0**
 - **Gateway** = **10.0.0.1**
 - **Primary DHCP Server** = **10.0.0.1**

2. 'Guest' **interface** - **VLAN ID (200)**
 - **VLAN Identifier** = **200**
 - **IP Address** = **10.1.0.100**
 - **Netmask** = **255.255.255.0**
 - **Gateway** = **10.1.0.1**
 - **Primary DHCP Server** = **10.1.0.1**

2.2 WLANs Tab

In this tab, we can assign the **logical interfaces** that we've created in the WLC Controllers tab to our **physical ports** on the WLC.

- We create a **WLAN** by going to the top right, selecting 'Create New' and pressing the 'Go' button.
- This will open a new page for us to create the **WLAN**, and it will ask us for these variables:
 - **Type**
 - **Profile Name**
 - **SSID**
 - **ID**

General Sub-Menu

- After applying the previous variables, we'll be sent to a new page where we can change the following:
 - **WLAN status** (Enabled or Disabled)
 - Assigning a **logical interface**.

Security Sub-Menu

- We can go into the 'Security' sub-menu and enable a few security functions, which are:
 1. **Layer 2**
 - **Layer 2 WPA+WPA2** security
 - **Pre-Shared Key (PSK)** authentication along with the **PSK** format (ASCII or HEX) and the key itself.
 2. **Layer 3**
 - Web Policy security
 - Web Policy type, there are multiple types:
 1. **Web Authentication**
 - ◆ After the wireless clients gets an IP address and tries to access a web page, they will have to enter a **username** and **password** to **authenticate**.
 2. **Web Passthrough**
 - ◆ Similar to the above, but no **username** or **password** are required. A warning or statement is displayed and the client simply has to agree to gain access to the Internet.
 3. **Conditional** and **Splash Page** Web Direct
 - ◆ They are similar options, but additionally require **802.1X Layer 2 authentication**.

QoS Sub-Menu

- Lastly, we can go to the 'QoS' sub-menu to set the **QoS** settings on our **WLC**.
- There are four options we can use:
 1. **Bronze** - (Background)
 2. **Silver** - (Best Effort)
 3. **Gold** - (Video)
 4. **Platinum** - (Voice)

WLAN Tab - Lab Configuration

For our current lab, we'll be setting up two WLANs:

1. 'Internal' **WLAN**
 - We've already set this up in the initial setup phase, so we only have to configure its settings.
 - **General Sub-Menu**
 - WLAN Status = **Enabled**
 - Logical Interface = **Internal**
 - **Security Sub-Menu**
 - Layer 2 Security = **WPA+WPA2**
 - Authentication Mode = **PSK**
 - **QoS Sub-Menu**
 - QoS Mode = **Silver** (Best Effort)

- 2. 'Guest' **WLAN**
 - Type = *WLAN*
 - Profile Name = *Guest*
 - SSID = *Guest*
 - ID = *2*
 - **General Sub-Menu**
 - WLAN Status = *Enabled*
 - Logical Interface = *Guest*
 - **Security Sub-Menu**
 - Layer 2 Security = *WPA+WPA2*
 - Authentication Mode = *PSK*
 - **QoS Sub-Menu**
 - QoS Mode = *Silver* (Best Effort)

2.3 Wireless Tab

This tab allows us to view APs that are currently connected to the WLC.

- Opening an **AP** will let us view its settings and configurations, including things like:
 - *AP Name*
 - *Location*
 - *AP Mode* (*Local*, *FlexConnect*, *Monitor*, [etc.](#))

2.4 Management Tab

This tab allows us to see information about the WLC operations.

- For example, you can see the current **SNMP** protocol being used, **HTTP** & **HTTPS** status, if **SSH** or **Telnet** are allowed or not, etc.

2.5 Security Tab

This tab gives us access over a wide range of security-related features. For the CCNA, it's enough for you to know about CPU ACLs.

- CPU **ACLs** are used to limit access to the CPU of the **WLC**. This limits which devices will be able to connect to the **WLC** via **SSH/Telnet**, **HTTP/HTTPS**, retrieve **SNMP** information from the **WLC**, etc.